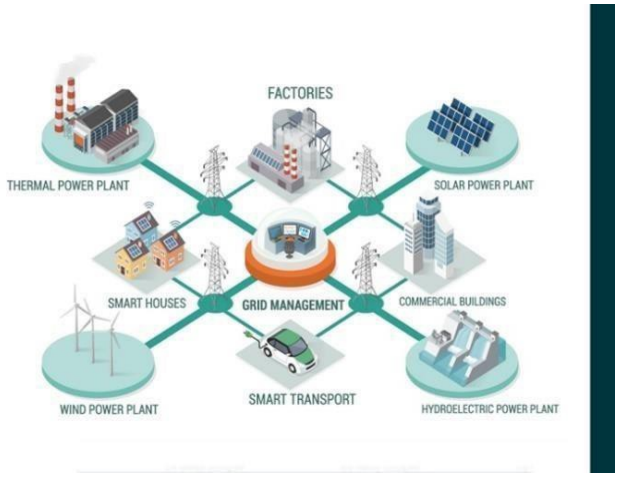
Technical Writeups/ Proposal



OWNER & EMPLOYER	**:**	**ASSAM ELECTRICITY GRID CORPORTION LIMITED**
BIDDER	**:**	**RBH SOLUTIONS PRIVATE LIMITED**
BID IDENTIFICATION NO	**:**	**AEGCL/MD/O&M(LAR)/SAS Upgrade Bilasipara GSS/2025/BID**
DUE DATE	**:**	**16-10-2025**
DOCUMENT REF NO.	**:**	**RBH/687/AEGCL/TB-01/2k25-26**
DATED	**:**	**14-10-2025**

SUBMITTED BY:

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RBH SOLUTIONS

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1. INTRODUCTION

1.1. Objective

This document is technical offer for "**Upgradation/Replacement of existing Substation Automation System at 132KV Bilasipara GSS (receiving end) and at 132KV Kokrajhar GSS (sending end), AEGCL**." against **Bid Identification No.: AEGCL/MD/O&M(LAR)/SAS Upgrade Bilasipara GSS/2025/BID** published by The Chief General Manager(O&M), LAR, AEGCL 1st Floor, Bijulee Bhawan, Paltan bazar Guwahati (Assam) 781001.

This document includes

- System Architecture
- Product Details
- Product Datasheets

1.2. Abbreviations

In this chapter the following shorten names (abbreviations) are used:

Employer	Assam Electricity Grid Corporation Limited
Bidder	RBH Solution Private Limited (RBH)
SAS	Substation Automation System
BCU	Bay Control Unit
HMI PC	Industrial Grade PC (IPC) hosting RsS SCADA /SAS Software including EMS & HMI Software.
Gateway PC	Industrial Grade PC (IPC) hosting RsS ICx Gateway and RsS ICx DR Gateway Software.
RsS SCADA Software	RBH RsS SCADA /SAS Software Solution Interfacing with RBH ICx Gateway over IEC 62443-2-4/ IEC61850 / IEC 60870-5-104 Protocols. Running on the IPC.
ICx Gateway	RBH ICx Gateway Solution for providing IEDs/BCUs (over IEC-61850) with SAS System and SLDC/Backup SLDC (over IEC 60870-5-104)
ICx DR Gateway	RBH ICx DR Gateway Solution for auto downloading facility for Numerical Relay Disturbance Recorder (DR's).
Ethernet Switch	Managed Ethernet Switch with required Fiber and Ethernet ports for interfacing with IEDs/ BCUs and other interfacing devices.

1.3. Tender Overview

Assam Electricity Grid Corporation Limited, a vibrant growth oriented Public Sector Company registered under 'Company Act, 1956'. It was formed out of restructured Assam State Electricity Board in 2003 and was notified as the State Transmission Utility (STU). Its core business is to efficiently transport electrical power from electrical power bulk heads to the distribution company networks in the state of Assam. Assam Electricity Grid Corporation Limited inherited 3862 circuit kms of EHV lines above 66 kV voltage class and 38 numbers of EHV sub-stations having a total transformation capacity of 1636.50 MVA at its birth in 2003.

Assam Electricity Grid Corporation Limited is playing a strategic role as it is the largest 'STU' in the NE region. It also understands its responsibility towards the entire North East India and is always extending a helping hand by way of transporting a fair share of power to the other sister states of the region.

Assam Electricity Grid Corporation Limited vows to ever strive till all transmission bottlenecks get eradicated. It pledges to deliver unrelenting brilliance in performance, deliver power efficiently but economically, show high safety standards, and is committed to respecting environmental and heritage issues.

1.4. RBH Solution Overview

RBH offered RsS SCADA and ICx Gateway-based Substation Automation System for d Upgradation/Replacement of existing Substation Automation System (SAS) that will enable the complete monitoring & control of all the bay equipment, Switchgear equipment with OWS/HMI placed at control rooms, with LIVE Data display and Historical data through integrated SQL Server.

RsS SCADA consists of Following Modules

- Data Acquisition Module
- Reports and Trending Module

RsS SCADA system, ICx Gateway & SAS software Platform, fully integrates with any vendor Energy Meters/ Protection Relays / Numerical Relays / BCUs as a single platform for the comprehensive management of the Substation and its associated infrastructure systems.

1.5. Document Overview

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- Scope of work, Compliance & General Terms in Chapter 2
- General Technical specification in Chapter 3
- System Overview and Architecture in Chapter 4
- RsS SCADA (SAS) & ICx Gateway Specifications in Chapter 5
- SCADA/SAS Functions in Chapter 6
- Quality Management System in Chapter 7

2. SCOPE, COMPLIANCE AND GENERAL TERMS

2.1. Overview

This chapter define the scope of work for Substation Automation System (SAS) as per the required features in tender documents and their compliance statements and general terms with reference to the tender documents:

2.2. Scope of Work for Substation Automation System (SAS)

The firm shall supply the subject material as per the detailed Technical Specifications and commissioning of the same indicated as below:

2.2.1. Upgradation/Replacement of existing Substation Automation System (SAS) and supply of spare equipment:

Sl.No.	Description of Item	Compliance
1.	Upgradation/Replacement of existing Substation Automation System (SAS) (Make: Alstom) at 132KV Bilasipara GSS, AEGCL and 132KV Kokrajhar GSS (Make: GE) as per BoQ and bid specification.	Complied
2.	Installation, testing, commissioning, training & handover of complete substation automation system	Complied
3.	Supply of licensed version of all necessary software with product keys	Complied
4.	Supply of all documentation related to the offered SAS	Complied
5.	Arrangement of onsite training of AEGCL personnel for operation and maintenance of the offered SAS	Complied
6.	Installation of all the necessary IED/DR analyzing software's shall be in the scope of the contracting firm	Complied
7.	Integration of existing IEDs, inverter along with GPS server and clock in the offered SAS shall be in the scope of the contracting firm	Complied
8.	Supply of Spare Equipments as per bid specification and BoQ	Complied
9.	The contracting firm must ensure compliance with the prevailing CEA cybersecurity norms and provide necessary updates in accordance with any changes to these norms during the contract period including defect liability period.	Complied
10.	Any upgradation of software shall be in the scope of contracting firm including license upgradation (if any)	Complied
11.	Arrangements of any permits required for transportation and movement of supplied materials shall be under the scope of the contractor.	Complied
12.	Transit insurance and insurance during storage at site till commissioning shall be in the scope of the contractor.	Complied

2.2.2. Annual Maintenance contract for a period of 10 (Ten) Years:

Sl.No.	Description of Item	Compliance
1.	The contracting firm should provide an Annual Maintenance Contract (AMC) of 10 (ten) years as per the BoQ and bid specification. Separate Agreement shall be done for Annual Maintenance contract as per bid terms and condition Appendix-6	Complied
2.	The AMC shall be of Non-Comprehensive Type	Complied
3.	Any upgradation of software shall be in the scope of contracting firm including license upgradation (if any).	Complied
4.	Integration of 10 Nos. of future bays (whenever required) shall be in the scope of the contracting firms as per BoQ	Complied
5.	Supply of softcopies of patch/upgrades with product keys etc. along with product documentation	Complied

2.3. Scope of supplies:

a) Supply and integration of all hardware and software components required for the upgraded SAS system.
b) Integration of all existing IEDs/relays into the new system without the use of gateways.
c) Ensuring the upgraded SAS architecture matches the existing system at 132kV Kokrajhar and Bilasipara GSS.
d) Configuration and parameterization of the system to ensure smooth communication between all devices.
e) Provision of necessary modems and interfaces for communication with SLDC.
f) Bay Control & Protection IEDs for control, monitoring and protection function.
g) Dual Redundant Gateways for remote communication (IEC 60870-5-101 & 104 protocols).
h) Redundant Power Supply cards for all IEDs, Gateways, and Network Equipment.
i) Supply of required cables, connectors, and accessories for LAN connectivity.
j) Factory Acceptance Test (FAT) and Site Acceptance Test (SAT) of the complete system.
k) Integration testing with existing SAS, SCADA, and SLDC.
l) Training to AEGCL personnel on operation and maintenance of the new SAS.
m) Supply of complete documentation, drawings, configuration files, and manuals.

2.4. Owner's Scope:

a) Existing IEDs/Relays and related protection/control wiring to be reused in the new SAS.
b) Existing SAS cabinets, control panels, and substation equipment not requiring modification.
c) Provision of communication link (PLCC / Fiber Optic) up to SLDC or STU/CTU wideband node.
d) Providing necessary space, power supply, and grounding facilities at the substation.
e) Coordination with SLDC for testing and approval of data exchange interfaces.
f) Approval of CERT-In empanelled auditor for VA/PT activities.
g) Providing the information regarding the existing system.
h) Providing Substation Drawings and protection schemes.
i) Providing access to the existing SAS servers, Engineering/DR PC & Remote Gateway.
j) Communication from the Substation to the Control Centres.
k) The Local Area Network for SAS system up to the Substation Switches for connecting the SAS servers, DR/Engineering PC, Remote Gateway.
l) Necessary shutdown/permits for testing of signals and integration.
m) Wiring of any signals required from the field up to the BCU, Protection relays & other IEDs.

n) Verification of the various configurations done during implementation stage and vendor has to support during verification.
o) Verification of local testing and End-to-End testing of the signals with the SLDC.
p) Coordinating integration with the Control centers.
q) Providing space in the existing control/Network panel for mounting of the supplied items.
r) Provision of source Power only. (Necessary wiring to be done by the vendor)
s) BCUs/Relays with Communication Protocol: IEC 61850 Compatible.
t) Providing necessary assistance to the contractor engineer working from the control centre end in case of troubleshooting and similar situations.
u) IEDs Engineering Software tools which are available at site.
v) Power source.

2.5. Exclusions from the Contractor's scope:

a) Supply, installation, or maintenance of communication links (PLCC / Fiber Optic) between substation and SLDC.
b) Any civil works or modification of substation building, cable trenches, or control room unless specifically mentioned.
c) Supply or modification of existing HV/MV/LV switchgear or primary equipment.
d) Upgradation or replacement of non-communicable legacy relays beyond the defined integration scope.
e) Supply & Installation of Panels and Furniture.
f) Supply of Cables, Cable Lying & Civil works.
g) Repair/ replacement of Existing Ethernet Switch/ Networking devices.
h) Checking of hardwiring of signals at Control & Relay panels and switchyard equipment for resolution of abnormal & suspect signals.
i) Defects resolution of issues related to communication & power cables, connectors, switches, routers & inverters which are not provided/supplied under the contract.

2.6. Annual Maintenance Contract (AMC)

2.6.1. Maintenance Support & Services

- Comprehensive maintenance of all supplied software and hardware to ensure a fully operational upgraded SAS system.
- Fault identification, repair/replacement of defective parts, and functional checks of the SAS system.
- Support services for integrating new bays, BCU, relays, and IEDs.
- Maintenance support includes 10 years of AMC post-Operational Acceptance of SAS System.
- Contractor to provide and arrange required tools, testing equipment, and materials for maintenance and system testing, including any special equipment needed.
- Maintain full operational SAS system and data reporting to control centers, ensuring all safety precautions.

2.6.2. Maintenance Activities

- Identification and resolution of SAS and Remote Gateway issues, such as suspect signals, unexecuted commands, non-reporting BCUs/IEDs, alarm mismatches, gateway/server hanging, time synchronization, and daily performance monitoring.
- Assisting site personnel for troubleshooting and rectifications.
- Incorporating database and SLD changes due to equipment or bay modifications.
- Emergency troubleshooting and breakdown maintenance, covering fault rectification, equipment replacement (e.g., servers, HMI, gateways) and fault analysis reporting.
- Conducting on-site visits for defect resolution, maintaining database backups, repairing/replacing gateways, and ensuring spare availability for system reliability.
- Performing system patch updates and integrating new signals into SAS servers, HMI, and Remote Gateways.

- Holding monthly meetings with the owner to address maintenance actions, submitting monthly and GAP analysis reports in finalized formats, and documenting all maintenance activities for each site.

2.6.3. Integration Activities Scope:

- The contractor is responsible for integrating new bays and equipment (e.g., IEDs, BCUs) into the SAS Server and Remote Gateway, ensuring the reporting of newly added data to remote control centers like Main/Backup STAMC /SLDC.
- Key integration services include:
- Database configuration in SAS Server, Remote Gateways, and Engineering/DR PC for DR acquisition.
- SCADA display setup in HMI.
- End-to-end testing between substations and control centers.
- Integration activities must follow a finalized checklist during detailed engineering, with costs for new bay integration included in the Annual Maintenance Charges (AMC).

2.6.4. Fault Reporting & Service Request:

- Faults will be reported by the Owner via email or phone, with the reporting time on phone used as the reference for resolution and response timelines. Details will follow the approved format.

2.6.5. Resolution & Response Time:

- The contractor will deploy an engineer to the site or resolve the issue remotely based on problem.
- Onsite visit scope is defined in AMC scope includes 2 nos. of Regular visits and 2 nos. of Emergency visits. For additional visit applicable man day and travelling cost shall be applicable.

2.6.6. Mandatory Spares Inventory:

- The contractor will maintain spare inventories at their expense to ensure system availability, using spares as needed without additional charges beyond maintenance costs.
- Suitable storage will be provided by the Owner at control centers.
- The contractor will determine additional spare requirements, maintain a minimum specified inventory, and replenish used spares within 45 days of award of project.
- Spare inventory will be enhanced if necessary to meet availability demands.

2.6.7. Exclusions from Contractor's scope in AMC Period:

The outage of data or function due to following reasons is excluded from contractor's downtime:

- Non reporting of data at control center end due to reason attributable to failure of tele communication network, Substation Router & Substation Switches provided by the Owner.
- Communication & Configuration issues of BCUs & Protection Relays.
- Repair of any Protection Relays & BCU.
- Relay firmware upgradation.
- Hard wiring of any signals from field or from Control room to the BCUs & Relays.
- Checking of Hardwiring for resolution of non-reporting & suspect signals.
- Defects resolution of issues related to communication & power cables, connectors, switches, routers & inverter which are not provided/supplied under the contract.
- Manual errors by Owner personnel.
- Power supply failure at Substation.
- Modifications and integration work required in the SCADA system at Control Centres.
- Shutdown for testing work as required.

2.6.8. Owner Responsibilities in AMC

The responsibilities of the owner during the maintenance period are as follows:

- IED/relay configuration files such as ICD, SCD etc. of new bays/elements to Contractor engineer.
- Provide signal list along with IEC 104 addresses.
- Shutdown for testing purpose.
- Arrange security passes/ permissions required by Contractor's engineer for required sites.
- Provide necessary assistance to the contractor engineer working from the control center end in case of troubleshooting and similar situations.
- Association of Owner Engineer for testing at Substation & Control Centre.
- Providing necessary assistance to the contractor engineer working from the control centre end in case of troubleshooting and similar situations.

2.7. General Terms & conditions:

2.7.1. Bid Validity

- The validity of the bid shall remain valid for a period of **180 days** from the date and time on which Price bids are opened.

2.7.2. Guarantee Required

The Contractor shall guarantee the availability of the complete Substation Automation System (SAS). The offer shall include detailed calculations demonstrating the guaranteed availability.

The Contractor will conduct an availability test on the entire SAS during pre-commissioning and commissioning to verify the reliability and integrity of all subsystems.

2.7.3. Completion Period

The completion period for fulfill all the works within five (5) months from the date of commencement of the contract.

2.7.4. Training:

The Contractor shall provide **on-site training** on the **operation and maintenance** of the SAS system to the Owner's personnel during the commissioning period. Additionally, the Contractor shall **submit the training modules** to the Employer for **prior approval** before conducting the training.

2.7.5. Spares

All consumables such as paper, cartridges shall be supplied by the contractor till the SAS is taken over by the Employer.

3. GENERAL TECHNICAL SPECIFICATION

3.1. Overview

This chapter define the General Technical Particulars of the offered items and associated specification and features complying as per the document.

3.2. Offered Bill of Material

Sl.No.	Description	Qty	Unit
	Supply & Service (Part-A)		
1.	**Supply: Software**		
1.1.	SAS software for substation monitoring and control (SAS License with 5000 tags or more shall be provided for this project)	4	Nos.
1.2.	EMS software for metering	2	Nos.
1.3.	Gateway software	4	Nos.
1.4.	DR/IED analyzing Software's	2	Job
2.	**Supply: Hardware**		
2.1.	**HMI-1 (Main)** Industrial grade Computer with minimum following specifications: Based on Intel Core 15, 12th Gen CPU, 16GB DDR4 Memory, 1*1TB 2.5" SATA SGD, DVD-Drive, 2*10/ 100/1000 mbps LAN Ethernet Ports onboard, 2*Serial, 4*USB, 1*VGA and 1*HDMI Port onboard, 1*USB Keyboard and Optical Mouse, 1*400-Watt Single Power Supply, Window-11 Pro, 64-bit OS, MS Office 2021, Antivirus (Make: Advantech/Technics or Equivalent)	2	Nos.
2.2.	**HMI-1 (Standby)** Industrial grade Computer with minimum following specifications: Based on Intel Core 15, 12th Gen CPU, 16GB DDR4 Memory, 1*1TB 2.5" SATA SGD, DVD-Drive, 2*10/ 100/1000 mbps LAN Ethernet Ports onboard, 2*Serial, 4*USB, 1*VGA and 1*HDMI Port onboard, 1*USB Keyboard and Optical Mouse, 1*400-Watt Single Power Supply, Window-11 Pro, 64-bit OS, MS Office 2021, Antivirus (Make: Advantech/Technics or Equivalent)	2	Nos.
2.3.	**Metering Workstation:** Industrial grade Computer with minimum following specifications: Based on Intel Core 15, 12th Gen CPU, 16GB DDR4 Memory, 1*1TB 2.5" SATA SGD, DVD-Drive, 2*10/ 100/1000 mbps LAN Ethernet Ports onboard, 2*Serial, 4*USB, 1*VGA and 1*HDMI Port onboard, 1*USB Keyboard and Optical Mouse, 1*400-Watt Single Power Supply, Window-11 Pro, 64-bit OS, MS Office 2021, Antivirus (Make: Advantech/Technics or Equivalent)	2	Nos.
2.4.	**DR PC:** Industrial grade Computer with minimum following specifications: Based on Intel Core 15, 12th Gen CPU, 16GB DDR4 Memory, 1*1TB 2.5" SATA SGD, DVD-Drive, 2*10/ 100/1000 mbps LAN Ethernet Ports onboard, 2*Serial, 4*USB, 1*VGA and 1*HDMI Port onboard, 1*USB Keyboard and Optical Mouse, 1*400-Watt Single Power Supply, Window-11 Pro, 64-bit OS, MS Office 2021, Antivirus (Make: Advantech/Technics or Equivalent)	2	Nos.

2.5.	**Gateway (Main)** Industrial grade Computer with minimum following specifications: Based on Intel Core 15, 12th Gen CPU, 16GB DDR4 Memory, 1*1TB 2.5″ SATA SGD, DVD-Drive, 4*10/ 100/1000 mbps LAN Ethernet Ports onboard, 2*Serial, 4*USB, 1*VGA and 1*HDMI Port onboard, 1*USB Keyboard and Optical Mouse, 1*400-Watt Single Power Supply, Window-11 Pro, 64-bit OS, MS Office 2021, Antivirus (Make: Advantech/Technics or Equivalent) OR Wall mounted redundant RTU Panel shall be provided as per specification	2	Nos.
2.6.	**Gateway (Standby):** Industrial grade Computer with minimum following specifications: Based on Intel Core 15, 12th Gen CPU, 16GB DDR4 Memory, 1*1TB 2.5″ SATA SGD, DVD-Drive, 4*10/ 100/1000 mbps LAN Ethernet Ports onboard, 2*Serial, 4*USB, 1*VGA and 1*HDMI Port onboard, 1*USB Keyboard and Optical Mouse, 1*400-Watt Single Power Supply, Window-11 Pro, 64-bit OS, MS Office 2021, Antivirus (Make: Advantech/Technics or Equivalent) OR Wall mounted redundant RTU Panel shall be provided as per specification.	2	Nos.
2.7.	32" LED Monitor, Resolution 1920*1080 (Make: Dell/Samsung or Equivalent)	8	Nos.
2.8.	A4 color Laser Printer (Make: HP/Canon/Xerox or Equivalent)	2	Nos.
2.9.	GPS clock with antenna and 50 mtr antenna cable (Make: Masibus or Equivalent)	2	Nos.
2.10.	3 KVA 1:1 Phase On-line UPS, (with Isolated Transformer) with Battery & Battery Rack (Make: Digiline or Equivalent)	4	Nos.
2.11.	2 Port RS485 To 1 Port Ethernet Converter, Metal Enclosure 24V DC Power input (Make: Data bridge or Equivalent)	2	Nos.
3.	**Freight and Insurance Charges**	2	Job
4.	**Services**		
4.1.	Engineering and configuration of complete Substation Automation System along with Factory acceptance Test	2	Job
4.2.	Installing, testing, commissioning, training and handover of complete Substation Automation System including integration of existing IEDs with the offered SAS	2s	Job
5.	**Spare Equipment (Part-B)**		
5.1.	Ethernet Switch ()	4	Nos.
5.2.	BCU spare (compatible with existing BCU MAKE: Hitachi, GE, Siemens)	4	Nos.
6.	**AMC**		
6.1.	Annual Maintenance Contract for Substation Automation System at 132KV Bilasipara GSS and 132KV Kokrajhar GSS along with scope for addition of 10 nos. of new feeder bays (whenever required) as per bid specification.	1	Job

3.3. General Specification of the offered items:

3.3.1. Technical Specification of Main & Standby HMI

Equipment	Specifications
Offered Make	Advantech/ Technics/ Eqvt.
Specifications	**19" 4U Rack Mountable, Industrial PC.** • Based on Intel Core i5 12th Gen. CPU. • 16GB DDR4 Memory. • 1*1TB 2.5" SATA SSD • 1*DVD-RW Drive. • 2*10/100/1000 mbps LAN Ethernet Onboard. • 2*Serial, 4*USB, • 1*HDMI & 1*VGA Ports onboard. • 1*USB keyboard and Optical Mouse. • 1*400-Watt Single Power Supply. • BIS Approved **Additional Peripheral** • 32inch, FHD, LED Monitor • Make Samsung/ Dell / Eqvt.
Operating system	Window-11 Pro, 64-bit OS.
Additional Software	MS Office 2021, Antivirus, RsS SAS Software

3.3.2. Technical Specification of Metering Workstation.

Equipment	Specifications
Offered Make	Advantech/ Technics/ Eqvt.
Specifications	**19" 4U Rack Mountable, Industrial PC.** • Based on Intel Core i5 12th Gen. CPU. • 16GB DDR4 Memory. • 1*1TB 2.5" SATA SSD • 1*DVD-RW Drive. • 2*10/100/1000 mbps LAN Ethernet Onboard. • 2*Serial, 4*USB, • 1*HDMI & 1*VGA Ports onboard. • 1*USB keyboard and Optical Mouse. • 1*400-Watt Single Power Supply. • BIS Approved **Additional Peripheral** • 32inch, FHD, LED Monitor • Make Samsung/ Dell / Eqvt.
Operating system	Window-11 Pro, 64-bit OS.
Additional Software	MS Office 2021, Antivirus, EMS Software.

3.3.3. Technical Specification of Main & Standby Gateway

Equipment	Specifications
Offered Make	Advantech/ Technics/ Eqvt.
Specifications	**19" 4U Rack Mountable, Industrial PC.** • Based on Intel Core i5 12th Gen. CPU. • 16GB DDR4 Memory. • 1*1TB 2.5" SATA SSD • 1*DVD-RW Drive. • 2*10/100/1000 mbps LAN Ethernet Onboard. • 2*Serial, 4*USB, • 1*HDMI & 1*VGA Ports onboard. • 1*USB keyboard and Optical Mouse. • 1*400-Watt Single Power Supply. • BIS Approved
Operating system	Window-11 Pro, 64-bit OS.
Additional Software	MS Office 2021, Antivirus, RsS ICx Gateway Software.

3.3.4. Technical Specification of DR PC.

Equipment	Specifications
Offered Make	Advantech/ Technics/ Eqvt.
Specifications	**19" 4U Rack Mountable, Industrial PC.** • Based on Intel Core i5 12th Gen. CPU. • 16GB DDR4 Memory. • 1*1TB 2.5" SATA SSD • 1*DVD-RW Drive. • 2*10/100/1000 mbps LAN Ethernet Onboard. • 2*Serial, 4*USB, • 1*HDMI & 1*VGA Ports onboard. • 1*USB keyboard and Optical Mouse. • 1*400-Watt Single Power Supply. • BIS Approved **Additional Peripheral** • 32inch, FHD, LED Monitor • Make Samsung/ Dell / Eqvt.
Operating system	Window-11 Pro, 64-bit OS.
Additional Software	MS Office 2021, Antivirus, DR/IED Analyzing Software

3.3.5. Technical Specification of A4 Printer.

Equipment	Specifications
Make	HP/Canon/Xerox/ Eqvt.

Specifications	• **Functions:** Print, Wireless
	• **Print Technology:** Laser
	• **Print Speed:** Black (A4): Up to 18 ppm; Color (A4): Up to 4 ppm
	• **Duty cycle (monthly, A4):** Up to 20,000 pages
	• **Media Sizes (Supported):** A4; A5; A6; B5
	• **Connectivity (standard):**
	  ○ Hi-Speed USB 2.0 port,
	  ○ Fast Ethernet 10/100Base-TX network port,

Quantity	2 Nos.

3.3.6. Technical Specification of GPS Clock

Equipment	Specifications
Make	Masibus/ Eqvt.
Specifications	**GPS Receiver** • Timing Accuracy: < 15 ns with GPS Receiver • Positioning Accuracy: < 10 m
	Interface and Configuration • Display: 6 digits, 0.56" (14mm) Seven Segment LED Display (Red) • Status LEDs: Power, 1PPS, Watchdog, GPS Locked **Power Supply** • 85-264V AC, 47 to 63 Hz / 120-300V DC **Physical** • Dimensions (mm) H x W x D: 144 X 72 X 140 **Environmental** Operating temperature: 0 to+55 °C
	Antenna • Type: Active L1. GPS, 30 dB gain • Antenna Cable: 50 Mtr

3.3.7. Technical Specification of 3KVA UPS

Equipment	Specifications
Make	Powertech/Microtek/ Elnova/ Eqvt.
Specifications	**3 KVA 1:1 Phase On-Line UPS** • Capacity (VA/W): 3000VA/2400W • Input Phase: Single phase with ground • Rated Input Voltage: 220Vac/230Vac/240Vac • Output Voltage: 220Vac (default) • Operating temperature Operating: 0~45°C, • Storage: -10ºC ~ 55ºC • Dimension (DxWxH): 469x191x339 mm **Battery & Accessories** • SMF Batteries 12V-120AH (6 Nos) • External Battery Charger, Battery Rack & Interlink Connector

3.3.8. Technical Specification of Serial Converter

Equipment	Specifications
Make	Databridge/ Eqvt.
Specifications	**ETHERNET (1 No.)** • Port Type: RJ45 Connector • Speed: 10 /100 M bps (Auto Detecting) **SERIAL PORT (2 Nos.)** • Port: 485/422 * 2 Port • RS485 Pins: Data+, Data- GND, Surge Protected • RS422 Pins: Rx+, Rx-, Tx+, Tx-, GND, Surge Protected • Speed: 300 bps to 921.6 Kbps, Parity: None, Odd, even

3.3.9. Offered Software Associated with the above Hardware's:

Sl. No.	Software Name
1.	Microsoft Windows 11 Professional
2.	RBH RsS EMS Software for Metering
3.	RBH RsS HMI
4.	RBH RsS SCADA/SAS Server Application
5.	RBH RsS Trends and Reporting
6.	RsS Historian
7.	RBH ICx Gateway Software
8.	RBH ICx DR/IED analyzing Software
9.	MS Office
10.	Antivirus

4. SYSTEM OVERVIEW AND ARCHITECTURE

4.1. Overview

This Chapter describes the Typical System Architecture for Substation Automation System offered solution which is complied with specification as asked in the tender document.

4.2. Substation Automation (SAS) Architecture

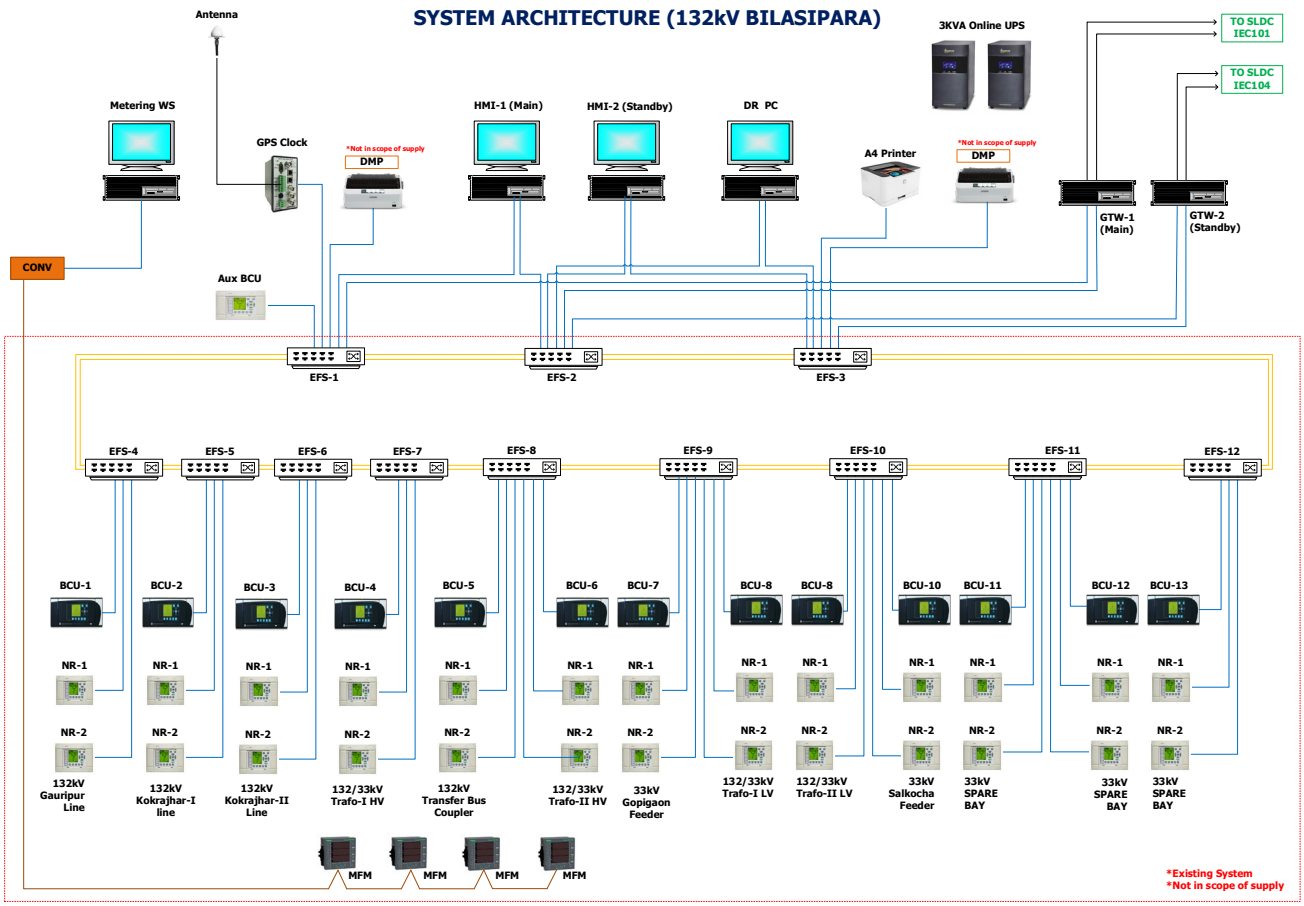


Figure 1: Substation-1 132KV Bilasipara GSS (receiving end)

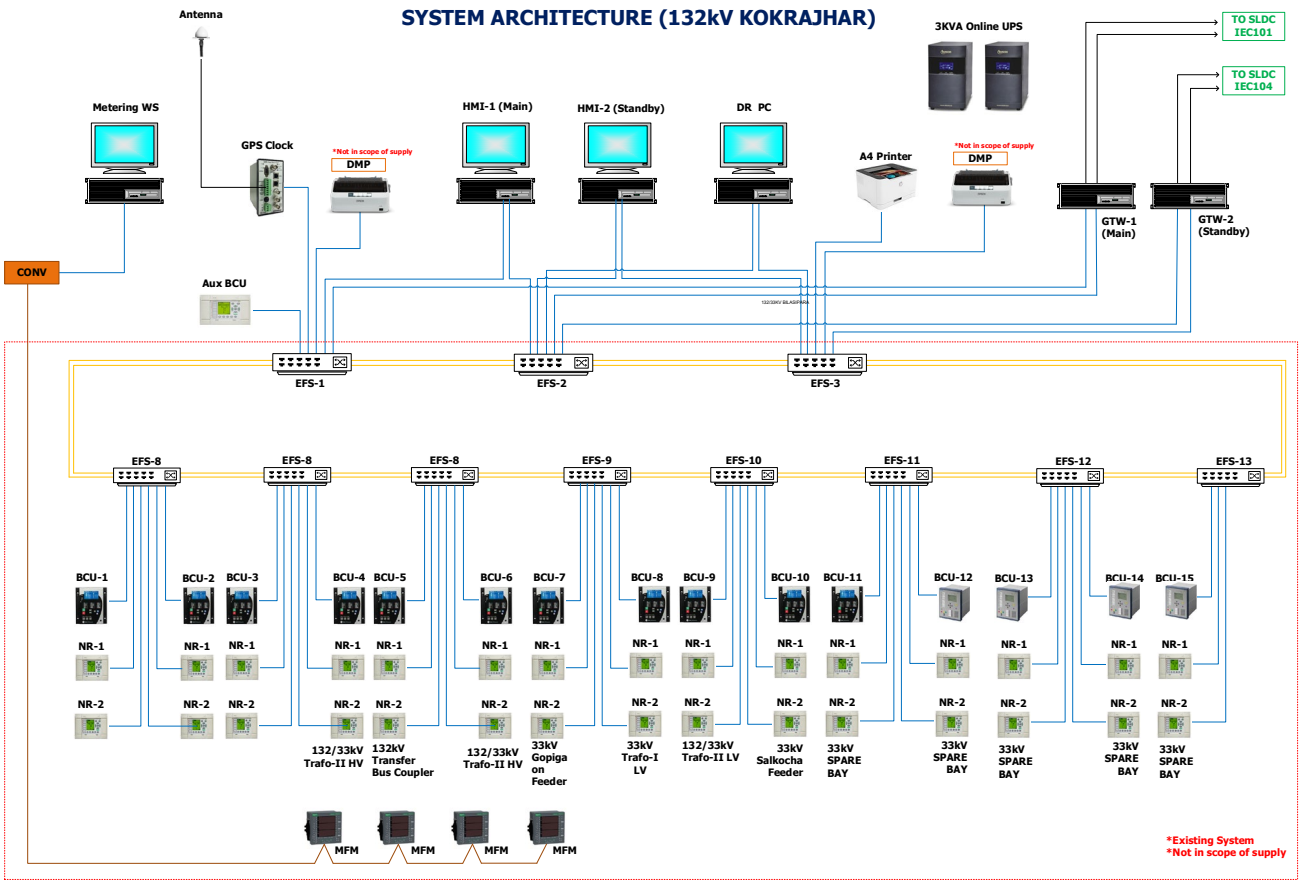


Figure 2: Substation-1 132KV Kokrajhar GSS (receiving end)

4.3. Systems Architecture Overview

Key architecture highlights are:

- There is Station Computer with redundant network links (as shown in the system architecture drawings)
- HMI System displays, real time data with facility for storage and analysis of Historical Data.
- This real time information can be the various electrical parameters or communication status of the IEDs/BCUs and System generated Reports and Trends.
- The RsS SCADA Application runs on Computers connected to RBH ICx Gateway through Ethernet network.
- Main Control Room:
  - Station HMI (Main & Stand-by)
  - Metering Workstation
  - DR PC
  - Gateway (Main & Stand-by)
  - LED Monitor
  - A4 Color Printer:
  - GPS Clock
  - UPS Inverter
  - Serial to Ethernet converter

4.4. System Introduction

4.4.1. Communication Networks

The offered Sub-Station Automation system (SAS) is based on following communication networks and data distributions. Summarized below are the levels of communication networks in the present architecture:

- **Station Network:** Communication between RBH RsS SCADA/ SAS and RBH ICx Gateway and communication between RBH ICx Gateway and SLDC over IEC 60870-5-104 Protocol.
- **Field Network:** Communication between IED's/ Relays/ BCUs and station networks on IEC 61850 Protocol. Communication between Meters and station networks on Modbus Protocol.

4.4.2. RBH ICx Gateway

The RBH ICx Gateway provides solutions for interfacing IEDs/BCUs and SAS System over various networks:

- RBH ICx Gateway interfaces with IEDs & BCUs over IEC 61850.
- RBH ICx Gateway interfaces with SLDC/Backup SLDC on IEC 60870-5-104 Protocol.
- RBH ICx Gateway interfaces with MFMs on Modbus Protocol.

4.4.3. Station Computers

The Station Computer allows the operator to supervise the electrical networks for various parameters through Gateway. Server PC and Gateway shall be installed at Control Room.

4.4.4. Station Network

The Ethernet network is the Local Area Network that links the RBH Station Computer (Server PCs & Gateway PCs) and Colour Printer for record printing including Dot Matrix Printer. The data to Station Computers will be on IEC 61850 / IEC 60870-5-104 protocol through network using LAN/ Ethernet. Station switches are connected in Ring Network.

4.4.5. Field Network

Field Network comprises of Network that links IEDs/BCUs on 61850 protocols through Ethernet network. The Network comprises of IEDs/BCUs on IEC 61850 connected to Ethernet Switches on optical fiber network and each switch shall be connected to Station switches over Ethernet Links.

4.4.6. RsS SCADA software

Station Computer (Main and Redundant) is built around RsS SCADA/SAS Application which employs a real time communication platform and database to update graphical object in real time. Station Computers will run RsS SCADA/SAS Application.

4.4.7. ICx Gateway and DR Software

Gateway PCs shall be built around ICx Gateway and ICx DR Gateway Application which employ data extraction from Relays/ BCUs & other communicable field devise and a real time disturbance recorder from Numerical Relays. Gateway PCs will run ICx Gateway Application and ICx DR Gateway Application.

5. RSS SCADA (SAS) SPECIFICATIONS

5.1. Overview

This chapter define the detailed technical specifications of the Substation Automation System (SAS) as per the required features in tender documents and their compliance statements

5.2. RsS SCADA/SAS Platform

5.2.1. Key System Functionalities

- Data Capture from BCU/ Relays/ Energy Meters or any Communicable Infrastructure.
- User defined High Resolution Graphics.
- Any type of Graphics can be created with multiple animation
- User defined Alarms / Events / Alert Colour Coding for the critical system states
- Energy Metering Data / Along with consumption report.
- Power System
- LDC Integration
- ABT
- KPI / Benchmarking

5.3. RsS SCADA/ SAS Modules

5.3.1. RsS EWS

- RsS EWS (Engineering Work Station): From the computer, user can on-line and off-line create, modify and suppress any configuration elements of the RsS solution.
- Creation, modification & suppression of any electrical, graphical and system data for Human Machine Interface.
- Import & export of data into MS Excel format or Binary Format

5.3.2. RsS DBM

- RsS DBM (Database Maintenance) Tool includes: Database Management: the database is created using pre-defined templates, their integrity is checked using expert rules, and they are downloaded into the SQL Database.
- Security Management: various password levels are defined & checked. Operator access Management can be offered through smart cards combining password and hardware key function if required.
- Device Operating Mode Management: this is used to start or stop a device

5.3.3. RsS Server

- RsS HMI SCADA Server: Core Service for communication, logging and control:
- Its main function is to handle all the Acquisition & Control and manage Database Management System.
- Communication with ICx DC Services for Field Data Acquisition.
- Server manages the supervision between all the devices and information is logged in Database System used by HMIs to animate the system supervision, indicating the place where (and when) a failure occurs.

5.3.4. RsS HMI

RsS HMI SCADA software: Operator Interface Software

- HMI are built using RsS HMI, which employs a real time database to update a graphical object in real time.
- Plant Electrical Views (SLD, Alarms, etc.), RsS Trends and Reports Viewer
- To views Trends, Time reports and object reports
- Reports can be directly exported to MS Excel format
- Allows reports and tends configuration settings management

5.4. ICx Gateway Modules

- ICx Platform based Gateway Firmware
- Firmware is responsible for following functions
  - Communication Protocols
  - Gateway Initialization, Running, Maintenance Modes
  - Interfacing with Gateway Configuration Tool
  - Gateway Configuration Loading, Validation and Operations accordingly

- ICx Gateway is based on Advantech Hardware
  - Based hardware with Processor, RAM, Hard disk & Linux based platform
  - Required Serial / LAN Interfaces as per Standard RS485 and Ethernet

- Communicate with Any make of Electrical Sensor / Meter over communication channels
- Communication with Control Centre over GPRS/ Network Modems / LAN / RS485 Serial Links

- Protocols
  - IEC 60870-5-101
  - IEC 60870-5-104
  - IEC 61850
  - OPC UA/ DA
  - Modbus TCP/IP
  - Modbus Serial

6. SCADA/ SAS FUNCTIONS

6.1. SCADA Coverage

The SCADA system used for this project shall be installed at following locations:

- Remote Location / Central Control Room
- Substation Control Room

6.2. SCADA Functionality

- The SCADA system provides real-time information in easy Graphical Representations allowing quick and accurate information availability.
- The SCADA System offered includes SCADA Server, Secure channel Gateways, and GPRS Router at the Substation, suitable for full control, metering, monitoring, supervision, storageof data, event logging and communication function of all Substations.
- The SCADA System Offered works on IEC60870-5-101 protocol.
- The SCADA System Offered is a computer-based system.
- Maintenance, modification or extension of component will not cause a shut off of the whole SCADA system. SCADA is suitable for extension in future for additional substation. During such event, normal operation of the existing Switchyard shall be Unaffected and system shall not require a shut down.

6.3. SCADA Maintenance

- Modification and evolution of the data bases of system requirements/configuration
  - SCADA System Maintenance and Configuration
  - Secure Communication Channel Maintenance and Configuration
  - Substation Router Maintenance and Configuration

- Parameterization of Devices
  - Substation Router allows transparent access to Meters / Devices Communication channels.
  - Tools to configure / manage Meters / Protection System (not in the scope, shall be provided by the suppliers respectively), installed on this terminal shall be able to access the remote device over this this transparent channel / tunnel.

6.4. Sequence of Event Recording

- The consignment of the changes of state of the Substation will be achievable directly on the SCADA. Following information will be available:
  - Date and state of Event / Alarm
  - Origin - name of the station, level of tension, name of the section, name of the equipment
  - Name of the object - Name of information (name of the logical entry)
  - Message of state/condition – State/condition clarifies logical entry (open/closed, beginning/end, etc.....)

6.5. Electrical Distribution Network Level and Supervision

6.5.1. Electrical Views

Animated single line diagrams of the electrical substations allow displaying the status of the different equipment with their associated measurements or status.

The Electrical synoptic screens on substation HMI include

- Overall view in which no control/command operation is not accessible,
- Detailed view of various electrical networks.

The following layers of electrical views are considered in this project.

- System Overview
- Single Line Diagram of each switchboard
- Detailed single line diagram of each incoming and outgoing feeder.

6.5.2. Supervision

- Presentation of single line topology: general sight of the substations, sight of each level of tension with different colors which will be chosen by the Client during the phase of approval.
- This presentation of topology will be dynamic depending on the states/conditions under tension or not under tension of single wires
- Sight of detail of each section. With the dynamic position of the circuit breakers, the value of measurements, the meters, the chart of alarms.
- Any spontaneous change, as the opening of a circuit breaker will be represented with a color and a specific state
- Presentation of the list of alarms, with buttons for Acknowledge and deletion
- Presentation of the curves of measurements in real time or extracts of archives. The invalid data will be indicated graphically.
- Impression of whole or part of the filed changes of state, alarms or the states of the various logical inputs/outputs with the generation of required report directly via the HMI and exportable towards offices applications (Excel)

6.5.3. Command and Control

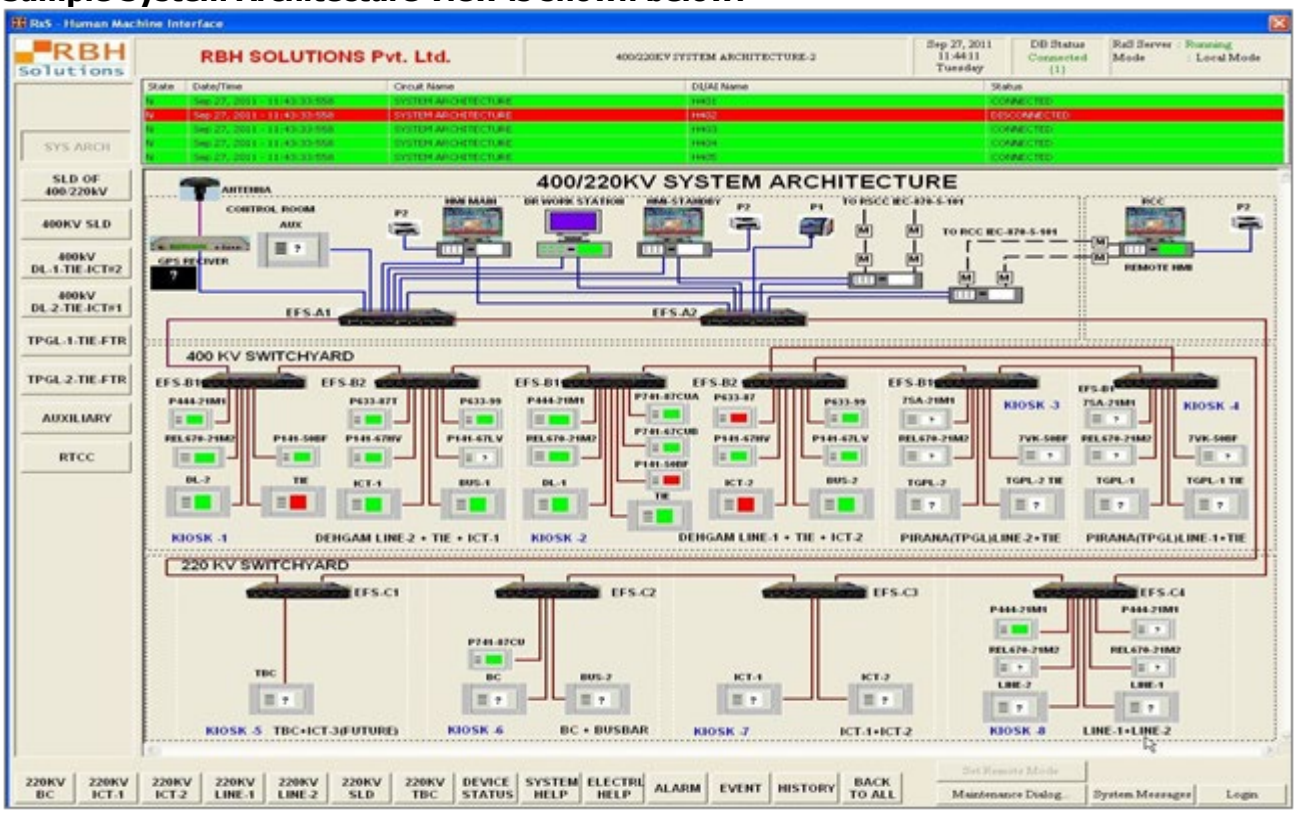
- Both Direct and SBO Command Supported
- Interlocking based operational control
- Interlock Bypass option
- Command Tagging / Blocking: Open, Close, Both
- Role based access to operators

6.5.4. System Views

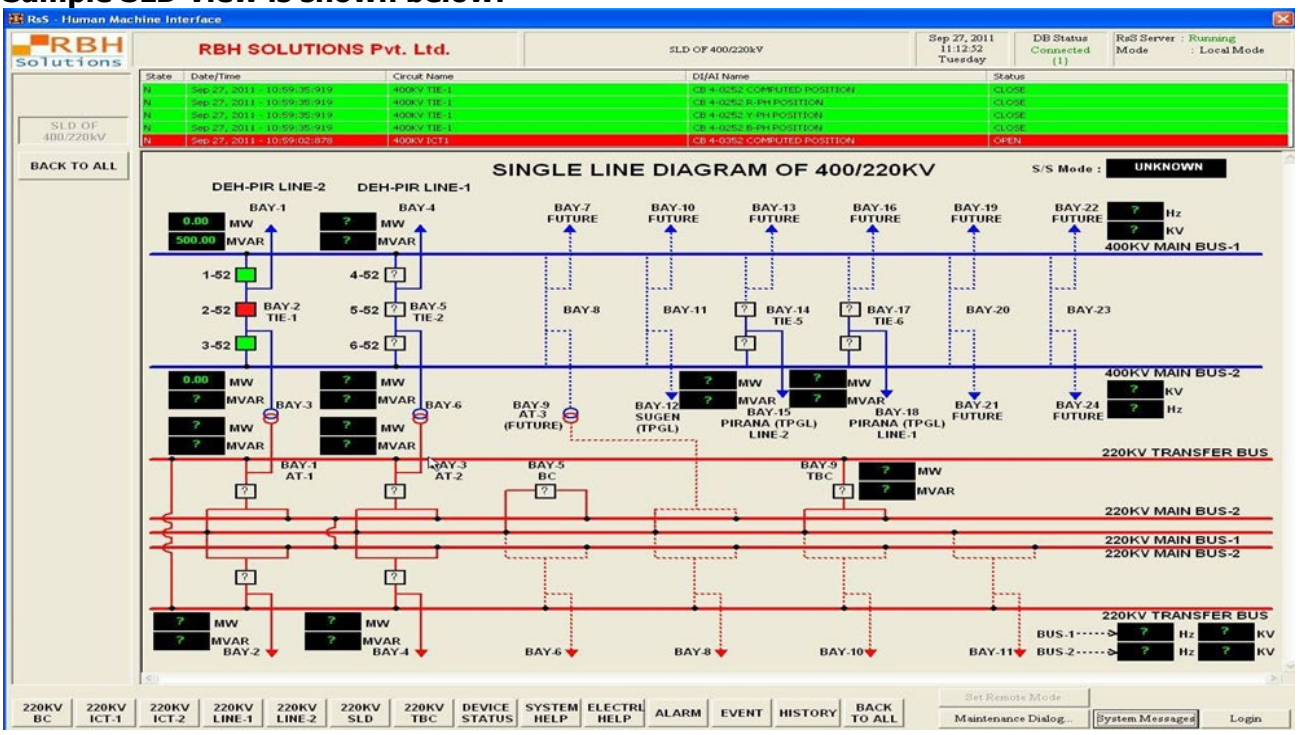
Animated diagrams of the electrical system allow displaying the status of the different devices. The following layers of System views are considered in this project.

- Overall view showing the architecture of the system and Connectivity Display
- SLDs
- Alarm Views
- Event Views
- Help View
- Object views to display all Tags of the selected Feeder / Switchboard

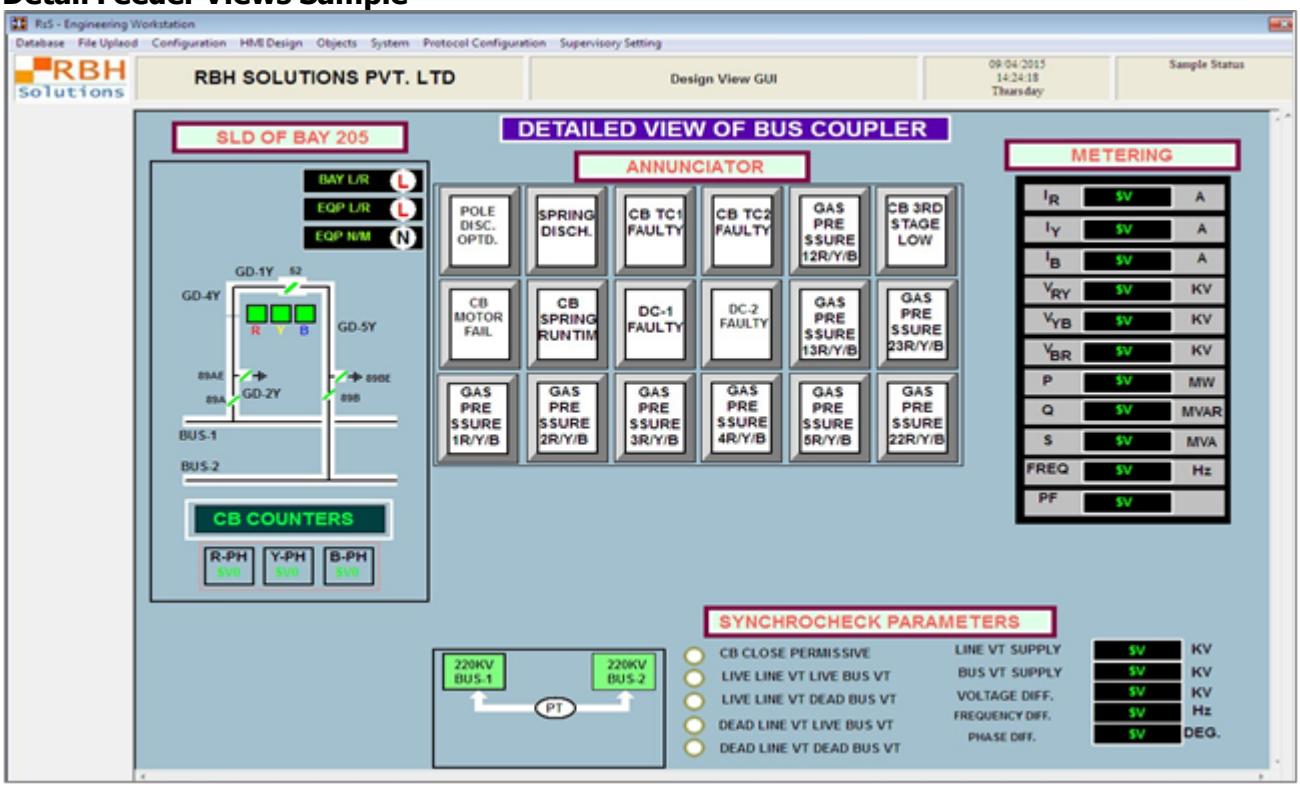
Sample System Architecture View is shown below:



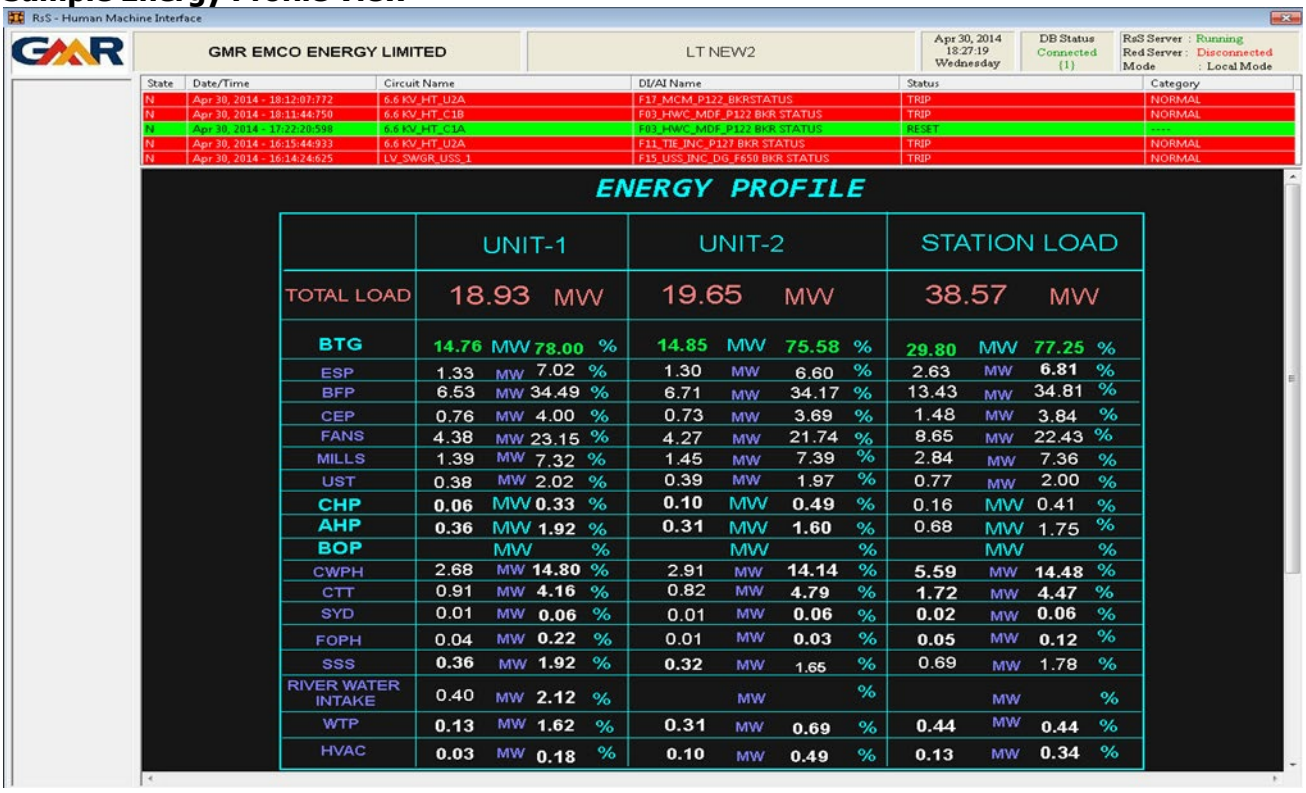
Sample SLD View is shown below:



Detail Feeder Views Sample



Sample Energy Profile View



6.6. COLOR DEFINITION

User can define any colour to any element at the SCADA HMI, we are proposing three colors red, green and Grey are used in the electrical views for animation. Red colour indicates that the feeder is charged or the breaker is closed, green colour indicates that the feeder is dead or the breaker is open and Grey Colour indicate don't believe it state.

Symbol	Double Point Input		State
	Input 1	Input 0	
	1	0	Open
	0	1	Closed
	1 0	1 0	DBI (Invalid state) Device in OPEN and CLOSE position

All analog values are displayed in blue colour in order to provide better localization at screen.

6.7. LOGGING

The logging function is used after an incident in order to understand the sequence of events and alarms in the electrical network. All events will be logged automatically and can be viewed by the operator.

6.7.1. The Alarm, Event and Status Viewer

By a single click on the corresponding button, the user chooses which list he wants to be displayed directly on the RsS HMI:

- the Alarm lists.
- the Event lists.
- the State lists.

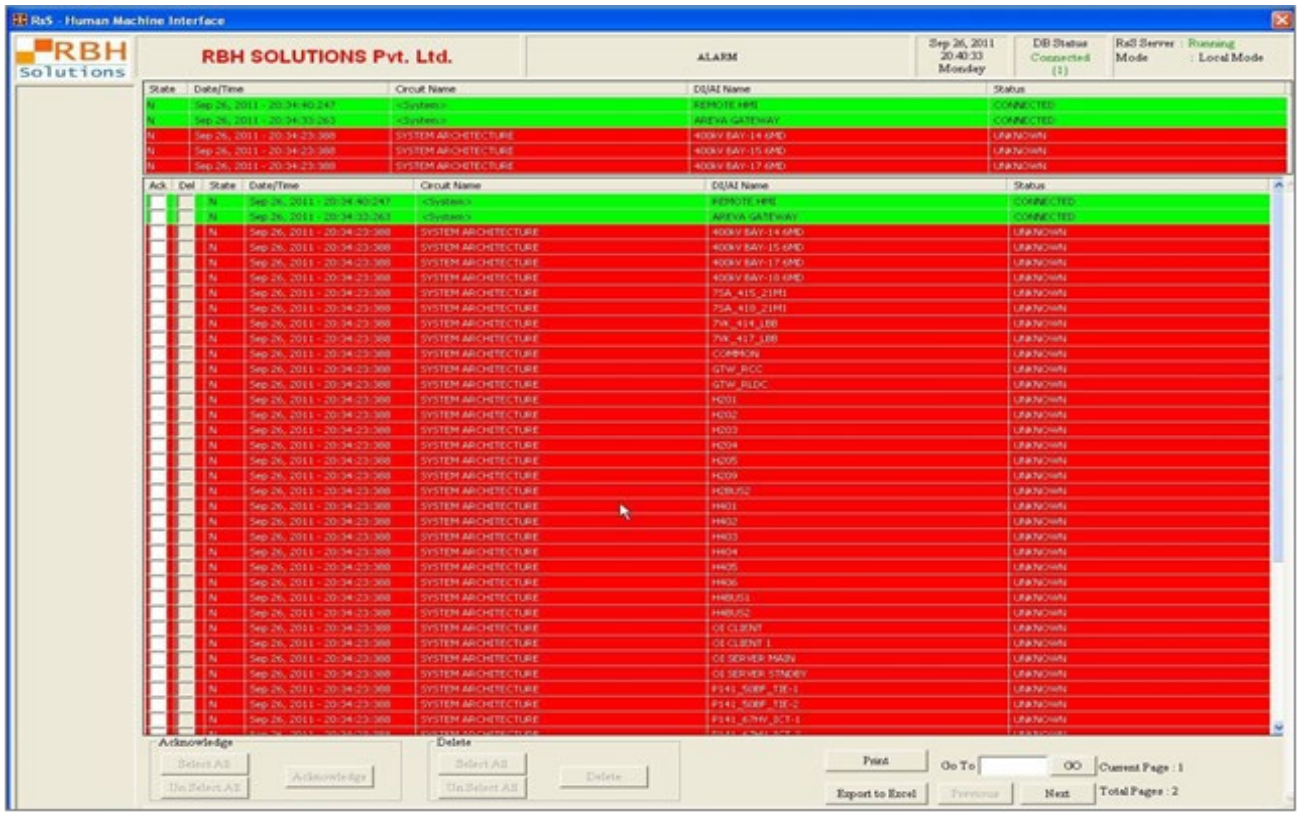
Each list is displayed with the actual number of cached pages and latest occurred information (alarms, new events, change of state) in addition to filtering and sorting facilities.

Dedicated buttons are used to:

- define filtering criteria
- print the list
- acknowledge the selected alarm(s), the alarm page, all alarms
- clear the selected alarm(s), the alarm page, all alarms

At electrical distribution network level, all the events generated by all the system devices can be configured to be event logged.

Alarm View



6.7.2. Archiving

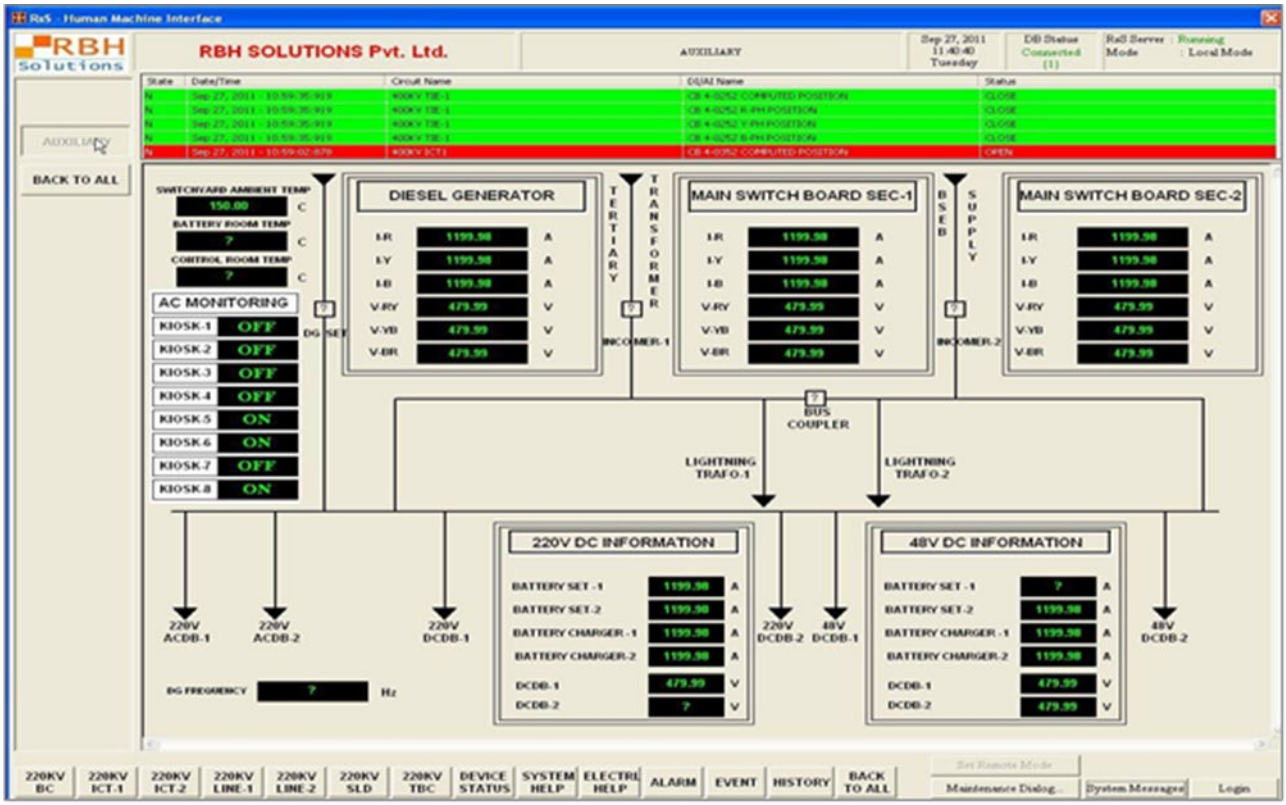
The stations operators will have a function of filing of telemetry which will make it possible to plot theload curves with durations from the day in the month, just as the edition of the reports/ratios of energy(E+, E, P, Q, S, U, I). This function will have:

- To store changes of state, measurements, and of all the actions of the operator

6.8. MEASUREMENTS

Measurements are made available from the Meters on the system Station Computers for eachsubstation.

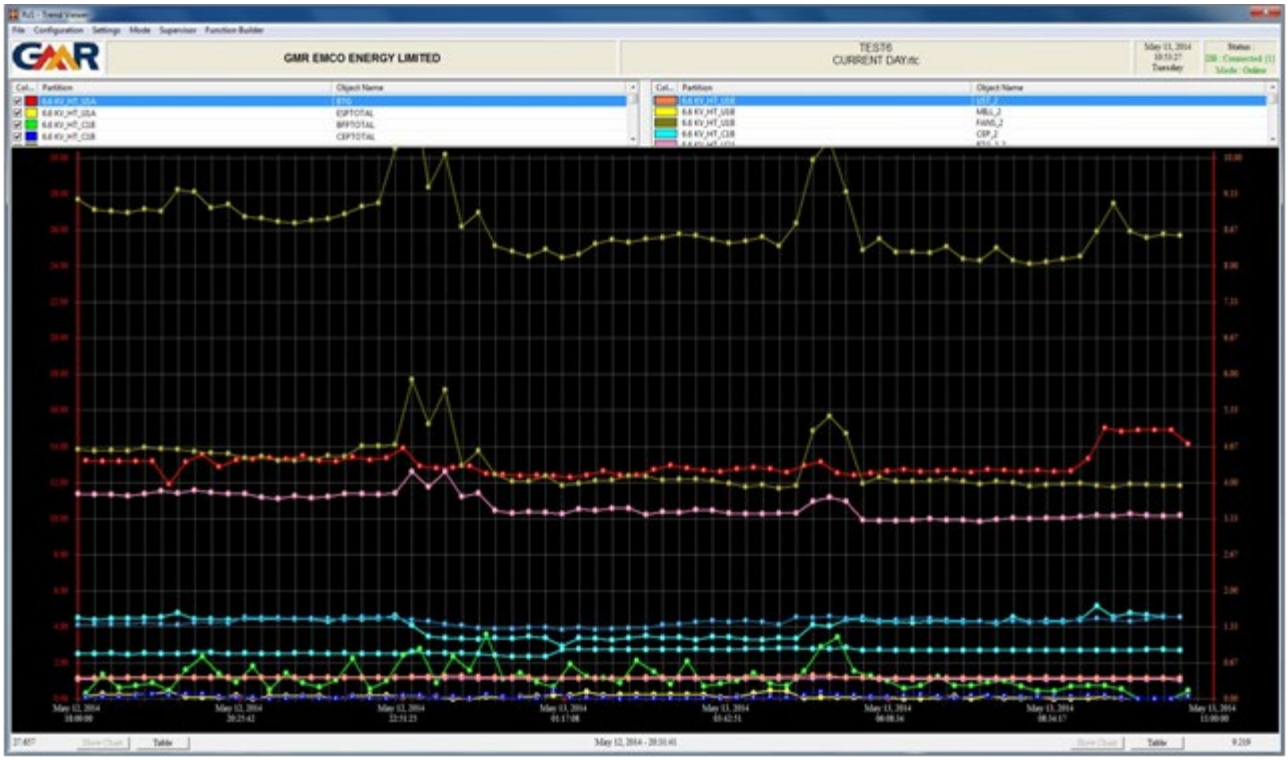
The main measurements are displayed in real time in the electrical views, distributed among theelectrical mimic. These are generally currents, voltages, frequency and power.



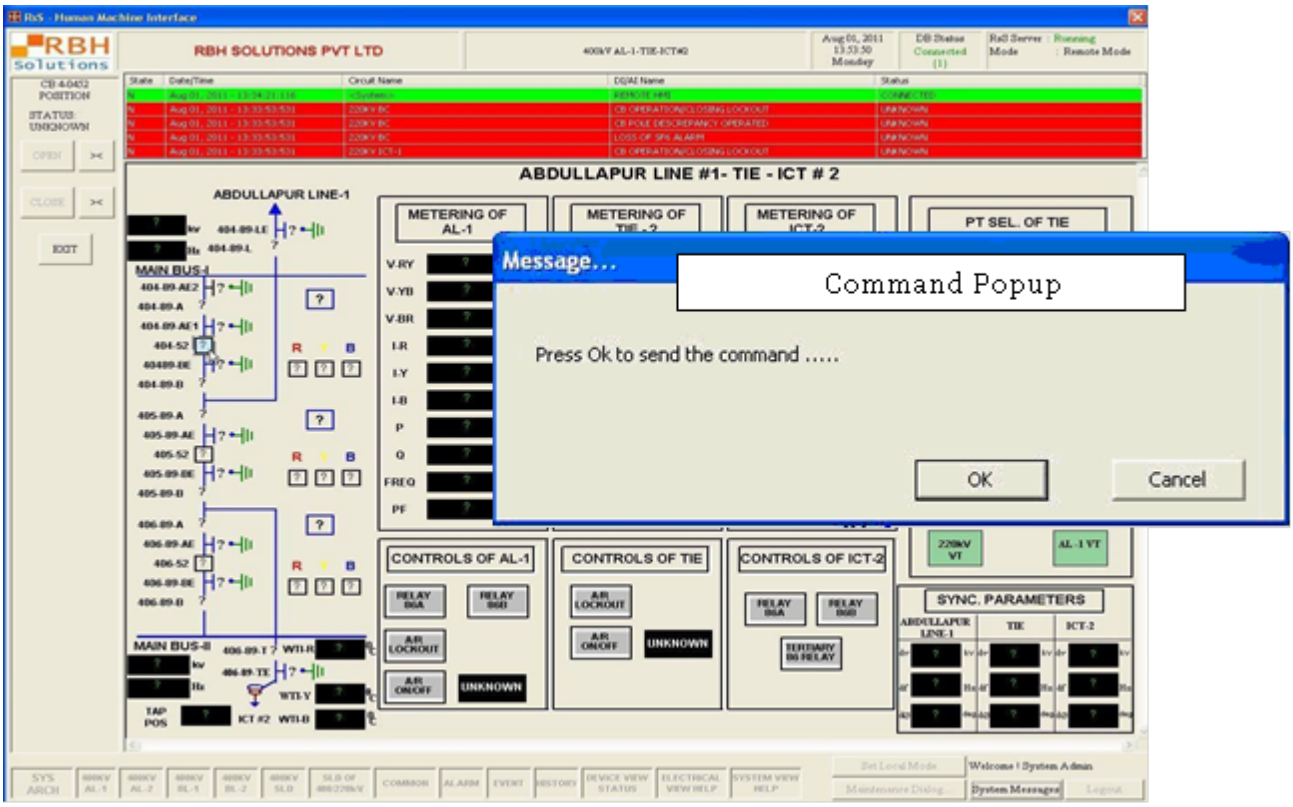
6.8.1. Measurement Trends

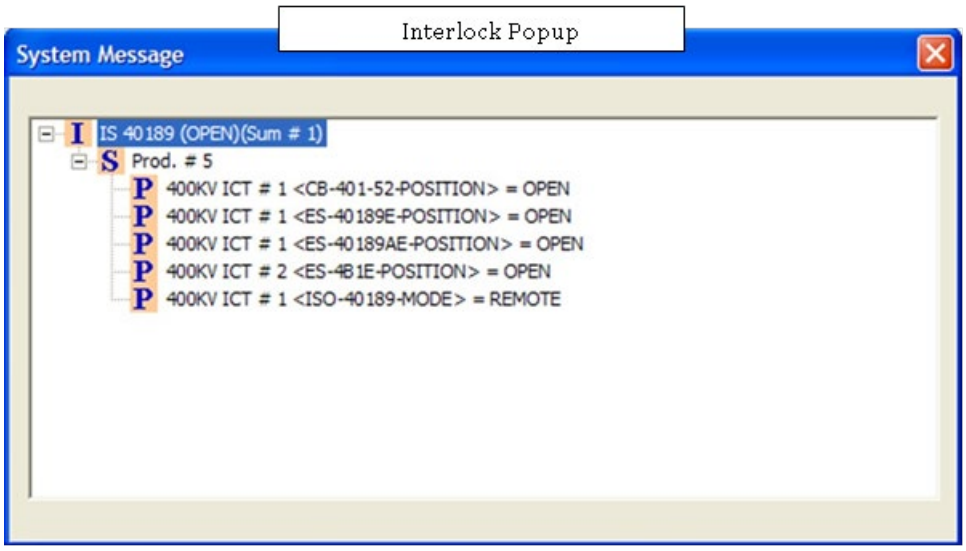
Measurement trends are available at the HMI. User selected analog inputs are recorded and displayed at regular intervals. Maximum 20 trends can be displayed at a time, and 200Sketches per configuration file can be displayed. Real-time trend with resolution of 1 Sec and Historical trends with resolution of 1 Minutes can be drawn. System will support tendsstorage capacity for 1 Year of data.

The Trend viewer of RsS Trends allows user to visualize real-time or historical evolution of any measurement configured to be logged. Data are available as line graph and application handles multiple points and supports user defined objects, time and colour attributes.



6.8.2. Control and Interlocking

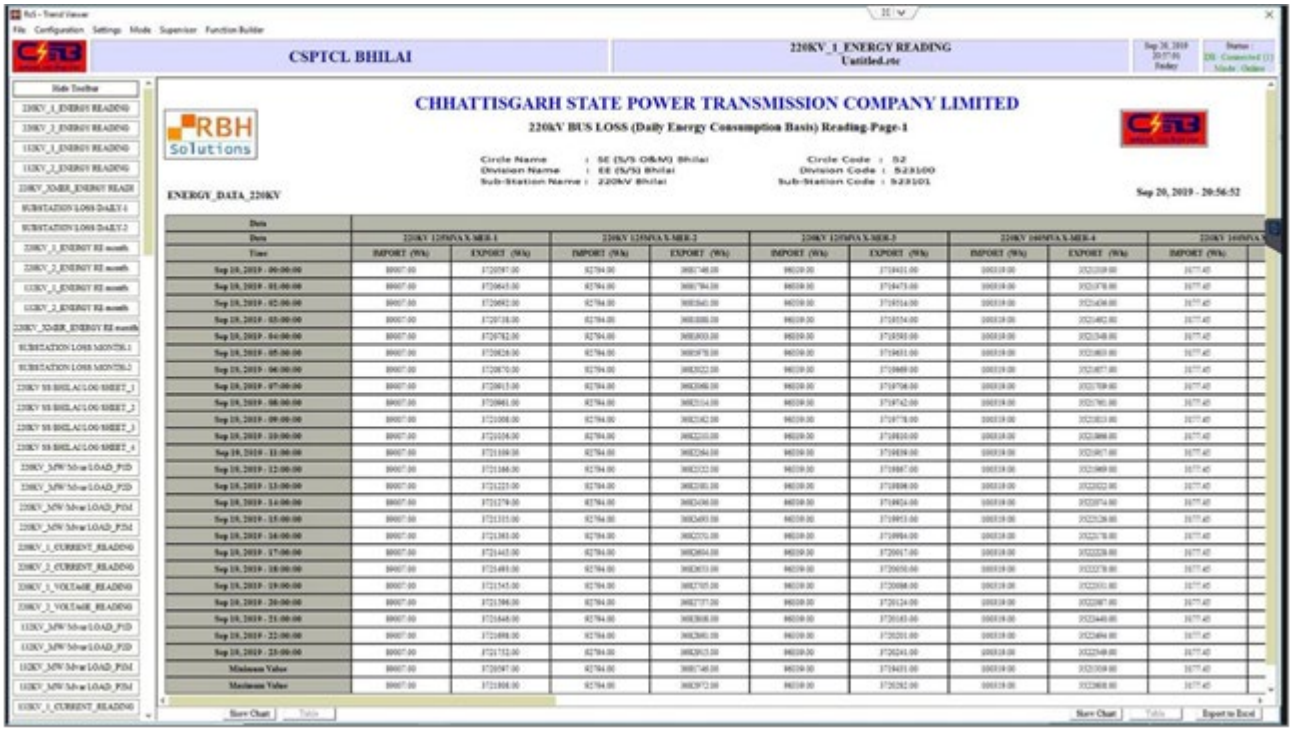


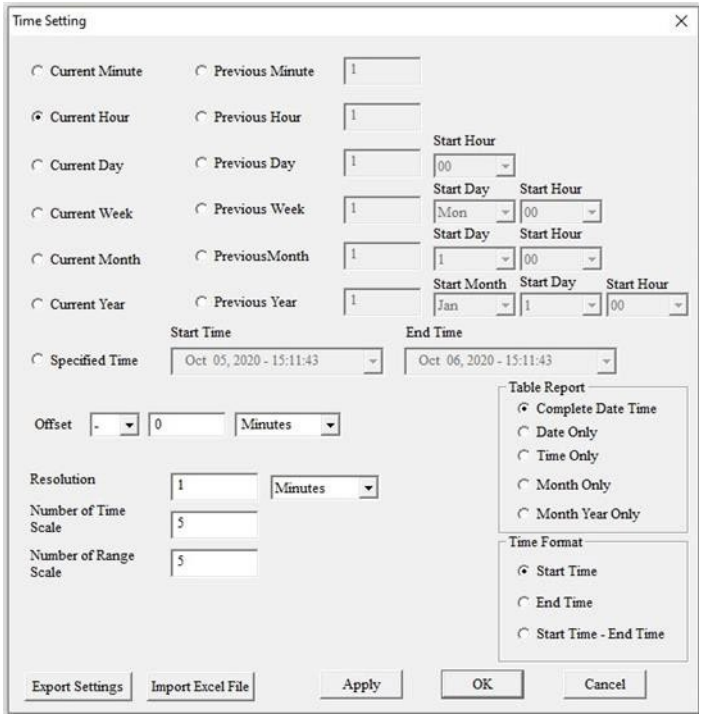


6.9. REPORTING

The reporting module of RsS Trends Application provides a mechanism for creating reports to show information collected and produced by RsS. It consists of a report format editor that allows the content and format of a report to be defined.

At runtime, a report generator will collect the relevant information and generate the report. Reports are saved in MS excel format with possible automatic deleting after a configurable duration. Facilities are provided for the viewing, edition and printing of reports





6.10. OPERATOR ACCESS MANAGEMENT

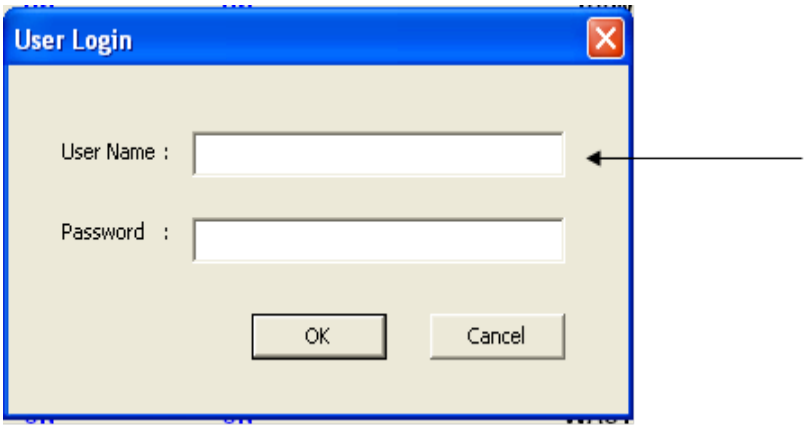
6.10.1. Operator Access Management

RsS HMI Application (RsS HMI) is an automation and protection control interface based on graphical presentation. Powerful graphical interface with windowing facilities and information management inDetail Views allows application to be examined with RsS HMI in detail.

The information is structured and presented to make monitoring simple, quick and efficient. Distributed control allows the system remains functioning in the event of a failure in any individual element / network.

6.10.2. Security Access

Access to RsS HMI is protected by user password. Different classifications of user are represented using user profiles. User can have different access rights to the various operational and engineering function of RsS HMI based on the type of profile they have been allocated. Each user can be configured to inherit a profile (from Administrator with full privileges to the default user with only ability to view part of the application).



6.10.3. Access Rights Profiles

The following user rights are available:

- Admin rights:
  - Database Administration
  - Stop the HMI application
- View rights: These rights are defined on a per mimic basis.
- Acknowledgement rights: alarm acknowledgement and clearing. These rights are defined on Bay point basis.

These user rights can be assigned to users. The user names and the associated password are created / deleted in EWS. Up to 256 user names can be defined. After a database switch, the users are automatically updated with these new access rights.

The password can be modified on-line by the user himself or a user with the "administrator" rights.

Access Level	Public	Operator	Engineer	Admin
Display System Status & View Screens	Yes	Yes	Yes	Yes
Acknowledge/Clear Alarms	No	Yes	Yes	Yes

6.11. Time Management

6.11.1. Time Synchronization

Time synchronization of Station Computers / IEDs is achieved though interfacing with Modular GPS Clock. Key modes of Time Synchronization are:

- From External GPS Clock over NTP / SNTP Protocol
- From Station Computer System Clock only when the External GPS Clock is not available

6.11.2. Time Distribution

RsS solution uses time stamping at source in order to achieve the required accuracy. All IEDs and SCADA System are synchronized with GTP Clock. Changes in state are then directly time stampedby the different IEDs. All IEDs are directly synchronized with GPS Clock and in case of GPS Clock Failure, Station Computer will act as backup NTP Server.

7. QUALITY MANAGEMENT SYSTEM

7.1. Quality Management Principles

RBH is ISO 9001:2015, Certified organization and focus on following quality management principles.

1) *Customer focus* – Being customer the key, Unique Central always focuses on striving to meet and exceed the current and future needs of their customers.
2) *Leadership* – Leader's inspiration and providing their people with adequate resources, training and empowerment to act within the set of well-defined responsibility. Also, leaders create a healthy environment by describing well defined policies considering employee, employer as well as company satisfaction.
3) *Involvement of People* – Involvement of people at all levels is very important for companies as well as individual's growth so that individuals accept ownership for problems and responsibility for solving them.
4) *Process Approach* – In order to achieve the desired result, activities and desired recourses are managed as the process, which allows for lower costs, improved cycle times, predictable results, and a focused approach.
5) *System Approach to Management* – Identifying, understanding, and managing company's interrelated processes as a system. This system provides more structured and integrated approach that recognizes the independences among processes and reduces cross functionalbarriers.
6) *Continual Improvement* – Continues improvement of products, processes and systems is thepermanent objective.
7) *Factual Approach to Decision Making* – All decisions made are based on accurate, reliable and analyzed data and facts. There is a balance between analysis based on facts or data andexperience and intuition for informed decisions.
8) *Manually beneficial supplier relationship* – An interdependent relationship exists between the customer and its suppliers. This relationship is mutually beneficial if this increases the abilityto create value for both parties. This value can support a long-term relationship that allows for polling expertise and recourses, while improving flexibility and speed in jointly responding to changing markets or customer needs.

7.2. Focused Standards

Focused Process/Product/Project Standards during Development

- Reliability
- Usability
- Performance
- Response
- Aesthetics
- Maintainability
- Reusability
- Conformance

7.3. Verification

Verification is defined as:

- The process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase.
- Did we build the product correctly?

At various stages of SDLC verification is done as following types of reviews

- Technical Reviews
- Business Reviews
- Management Reviews

7.4. Validation

Validation is defined as:

- The process of evaluating a system or component during or at the end of the development process to determine whether it satisfies specified requirements.

- Does the product do what it should?"

7.5. VEE Model

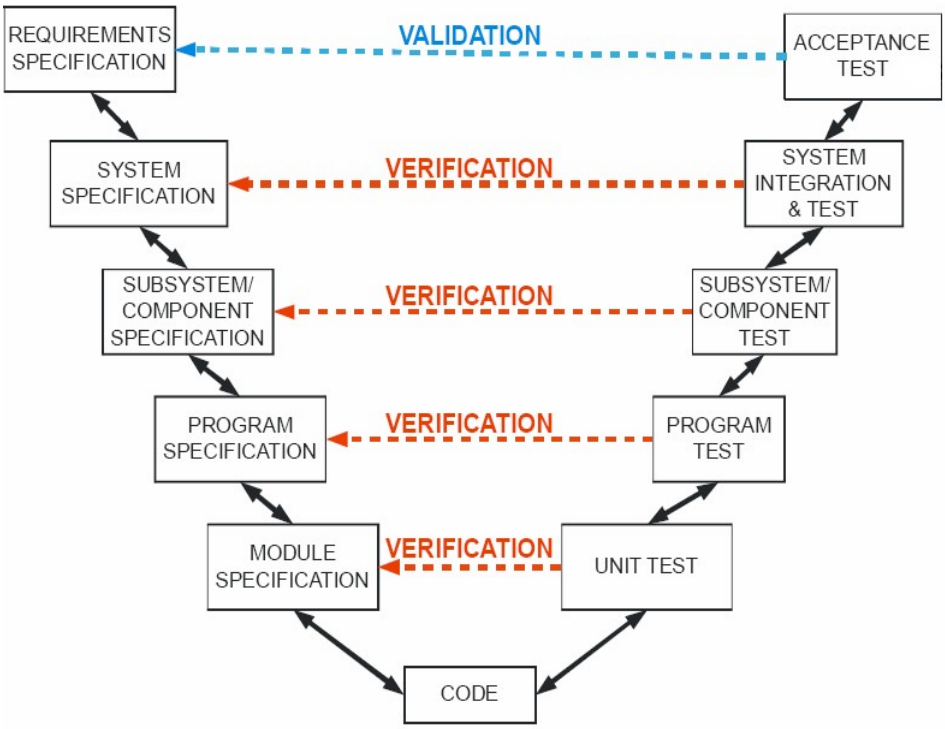


Fig 1. VEE Model

7.6. Objectives of VEE Model

The V-Model provides guidance for the planning and realization of projects. The following objectives are intended to be achieved by a project execution:

- *Minimization of Project Risks*: The V-Model improves project transparency and project control by specifying standardized approaches and describing the corresponding results and responsible roles. It permits an early recognition of planning deviations and risks and improves process management, thus reducing the project risk.

- *Improvement and Guarantee of Quality:* As a standardized process model, the V-Model ensures that the results to be provided are complete and have the desired quality. Defined interim results can be checked at an early stage. Uniform product contents will improve readability, understandability and verifiability.

- *Reduction of Total Cost over the Entire Project and System Life Cycle:* The effort for the development, production, operation and maintenance of a system can be calculated, estimated and controlled in a transparent manner by applying a standardized process model. The results obtained are uniform and easily retraced. This reduces the acquirer's dependency on the supplier and the effort for subsequent activities and projects.

- *Improvement of Communication between all Stakeholders:* The standardized and uniform description of all relevant elements and terms is the basis for the mutual understanding between all stakeholders. Thus, the frictional loss between user, acquirer, supplier and developer is reduced.

7.7. Product Support

Our Post-Sales Repair Service is equal in importance to our Design and Manufacturing. The service represents our commitment to provide comprehensive technical support after delivery of products/ Solutions. Through dedicated Customer Support team, our clients can get required technical support and repair services along with a stringent, dependable quality standard.

Key support system timelines are:

- Phone Support: 1 Hr
- WhatsApp Support: 1 Hr
- Email Support: 1 Day
- Remote Login Support: 1 Day
- Onsite Support: 2 Day